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NASAL CATHETER ASSEMBLY AND METHODS OF USE

Abstract

A nasal catheter assembly includes a balloon catheter that has a lumen and a first balloon in communication with the lumen. A hub is connected to the balloon catheter and defines an inlet port. The hub includes a housing that defines a plurality of ridges extending outwardly. The assembly also includes an absorbent member that has an outer surface configured to absorb a liquid and an inner surface that defines a cavity. A washer is moveably disposed over the hub housing and is disposed within the cavity of the absorbent member such that the washer engages the inner surface of the absorbent member. A nut is movably disposed over the hub housing and has at least one finger sequentially engageable with the plurality of ridges for incrementally moving and securing the washer and absorbent member at any one of a plurality of positions along the longitudinal axis of the housing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of the filing date of U.S. Provisional Application No. 63/560,824, filed Mar. 4, 2024, the disclosure of which is hereby incorporated herein by reference.

BACKGROUND

[0002] Epistaxis (nosebleed) is a very common problem in the general public. Epistaxis can be divided into two types: anterior epistaxis (i.e., nosebleed originating at the front of the nasal cavity) and posterior epistaxis (i.e., nosebleed originating at the rear of the nasal cavity). Anterior epistaxis, which usually occurs at a collection of blood vessels at the nasal septum, is far more common than posterior epistaxis and can be more easily controlled because the blood vessels that usually cause this bleeding are small in size, superficially located, and easily compressible via internal nostril packing and/or external nose pinching. On the other hand, posterior epistaxis is more difficult to control because it is generally caused by a larger blood vessel (i.e., sphenopalatine artery) which is located in a region deep within the nasal cavity making it difficult to reach and compress via direct pressure.

[0003] In addition to being more difficult to control than anterior epistaxis, posterior epistaxis can pose serious health risks. Typically, patients who suffer from posterior epistaxis are elderly individuals who suffer from hypertension and may be on blood thinning medication which can significantly reduce the ability of blood to clot and bleeding to stop. Thus, these individuals are at high risk of excessive blood loss that can lead to heart attack, stroke, pneumonia, and even death. The goal is to stop the bleeding as soon as possible. This can typically be accomplished by an experienced Otolaryngologist (Ear, Nose and Throat Specialist). However, in many situations the only healthcare provider available may be an EMT (Emergency Medical Technician), paramedic, nurse, or an emergency room physician who may not have the training and experience to appropriately address posterior epistaxis with the currently available devices which are not user friendly and/or efficient. Therefore, further improvements are desirable.

BRIEF SUMMARY OF THE DISCLOSURE

[0004] In one aspect of the present disclosure, a nasal catheter assembly includes a balloon catheter that has a first end, a second end, a lumen extending through the first end toward the second end, and a first balloon in communication with the lumen such that fluid flowing through the lumen fills and expands the first balloon. The assembly also includes a hub connected to the first end of the balloon catheter. The hub defines an inlet port in communication with the lumen of the balloon catheter such that fluid introduced through the inlet port is directed into the lumen. The hub includes a hub housing that defines a plurality of ridges that extend outwardly therefrom and transverse to a longitudinal axis of the hub. Also included in the assembly is an absorbent member that has an outer surface that is configured to absorb a liquid and an inner surface that defines a cavity. A washer is moveably disposed over the hub housing and is disposed within the cavity of the absorbent member such that the washer engages the inner surface of the absorbent member. Further, a nut is moveably disposed over the hub housing and has at least one finger sequentially engageable with the plurality of ridges for incrementally moving and securing the washer and absorbent member at any one of a plurality of positions along the longitudinal axis of the housing.

[0005] Additionally, the absorbent member may be a conical sponge that may have an opening that extends therethrough and which may communicate with the cavity. The balloon catheter may extend through the opening of the absorbent member. The inner surface of the absorbent member may taper inwardly from a first end toward a second end thereof, and the washer may include an outer surface that has a taper corresponding to the taper of the inner surface of the absorbent member such that advancing the washer within the cavity of the absorbent member applies an outward radial force on the absorbent member. The washer may include a convex surface, and the nut may include a concave surface engageable with the convex surface of the washer. The washer and hub may include corresponding flats which may permit slidable translation of the washer over the hub and may prohibit rotation of the washer relative to the hub. Also, at least a portion of the nut may be disposed within the cavity of the absorbent member and may engage the washer. At least one finger of the nut may be a flexible finger configured to permit axial translation of the nut in a first axial direction and resist axial translation in an opposite second axial direction when engaging any one of the ridges of the plurality of ridges. The plurality of ridges may be defined by at least one thread helically extending about the hub housing, and rotating the nut about the longitudinal axis of the hub housing in a first rotational direction may move the nut in the second axial direction.

[0006] Furthermore, the hub may include a first connector member and a second connector member which may be connected to the first connector member. The first connector member may define the inlet port and may include a one-way valve therein. The second connector member may define a connector end connected to the first end of the balloon catheter. The hub housing may include first and second housing portions disposed over at least a portion of each of the first and second connector members. The hub may further include a housing ring that may be disposed over at least a portion of the first and second housing portions and may secure the first and second housing portions together. The balloon catheter may further include a second balloon in communication with the lumen such that fluid flowing through the lumen may also fill and expand the second balloon.

[0007] In another aspect of the present disclosure, a nasal catheter assembly includes a balloon catheter that may have a first end, a second end, a lumen extending through the first end toward the second end, and a first balloon in communication with the lumen such that fluid flowing through the lumen fills and expands the first balloon. A hub is connected to the first end of the balloon catheter and defines an inlet port in communication with the lumen of the balloon catheter such that fluid introduced through the inlet port is directed into the lumen. The hub includes a hub housing that defines a plurality of ridges extending outwardly therefrom and transverse to a longitudinal axis of the hub. The assembly also includes an absorbent member that has a conically tapered outer surface that is configured to absorb a liquid and an inner surface that defines a cavity. A nut is movably disposed over the hub housing and is at least partially disposed within the cavity of the absorbent member. The nut has at least one finger that is sequentially engageable with the plurality of ridges for incrementally moving and securing the washer and absorbent member at any one of a plurality of positions along the longitudinal axis of the housing.

[0008] In a further aspect of the present disclosure, a nasal catheter assembly includes a balloon catheter that has a first catheter that defines a lumen, a second catheter disposed over the first catheter, and first and second balloons disposed over the first and second catheters. The first and second catheters each have a plurality of transverse openings that extend transverse to respective longitudinal axes thereof and are arranged relative to each other such that fluid flowing from the lumen and from the transverse openings fills and expands the first balloon and then the second balloon. A hub is connected to the balloon catheter and defines an inlet port in communication with the lumen of the balloon catheter such that fluid introduced through the inlet port is directed into the lumen. An absorbent member is slidably disposed over the hub.

[0009] Additionally, the assembly may also include a washer disposed within the absorbent

member, which may be a conical sponge. The washer may be slidably disposed over the hub and rotationally constrained relative thereto. The first balloon may extend over an end of the first catheter, and the lumen of the first catheter may extend through the end. The second catheter may have an outer surface that may define a plurality of longitudinally extending ribs, and an inner surface that may define a plurality of longitudinally extending ribs. The ribs of the inner surface may engage the first catheter, and the ribs of the outer surface may interface with the first and second balloons. At least one pair of transverse openings of the first and second catheters that feed fluid to the first balloon may be aligned, and each of the transverse openings of the first and second catheters feeding fluid to the second balloon may non-aligned.

[0010] In yet another aspect of the present disclosure, a nasal catheter assembly includes a balloon catheter that has a pre-curve defining a center of curvature and first and second balloons. The first and second balloons are each configured to expand eccentrically away from the center of curvature when fluid is introduced to the first and second balloons through the balloon catheter. A hub is connected to the balloon catheter and defines a fluid inlet port in communication with the balloon catheter. Also, an absorbent member is slidably disposed over the hub.

[0011] Additionally, the balloon catheter may include a first catheter that may define a lumen of the balloon catheter and a second catheter that may be disposed over at least a portion of the first catheter. A line of adhesive may secure at least a portion of the second balloon to the second catheter. The first catheter may be made from a first material, the second catheter may be made from a second material, and the second material may be more flexible than the first material. The first material may be polyetheretherketone, and the second material may be silicone.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. 1 is a schematic view of an exemplary embodiment of a nasal catheter system.

[0013] FIG. 2 is an exploded view of an exemplary embodiment of a nasal catheter assembly of the system of FIG. 1.

[0014] FIG. 3A is an elevational view of an exemplary embodiment of a balloon catheter of the nasal catheter assembly of FIG. 2.

[0015] FIG. 3B is a cross-sectional view of the balloon catheter taken along a line B-B of FIG. 3A.

[0016] FIG. 3C is an elevational view of the balloon catheter of FIG. 3A in a pre-curved configuration.

[0017] FIG. 4A is an elevational view of an exemplary embodiment of a stylet of the balloon catheter of FIG. 3A.

[0018] FIG. 4B is an elevational view of the stylet of FIG. 4A rotated 90 degrees relative to FIG. 4A.

[0019] FIG. 4C is an end view of the stylet of FIG. 4A.

[0020] FIG. 5A is an elevational view of an exemplary embodiment of a backbone of the balloon catheter of FIG. 3A.

[0021] FIG. 5B is an elevational view of the backbone of FIG. 5A rotated 90 degrees relative to FIG. 5A.

[0022] FIG. 5C is an end view of the backbone of FIG. 5A.

[0023] FIG. 6 is an elevational view an exemplary embodiment of a hub of the nasal catheter assembly of FIG. 2.

[0024] FIG. 7 is an elevational view of an exemplary embodiment of a first connector member of the hub of FIG. 6.

[0025] FIG. 8 is an elevational view of an exemplary embodiment of a second connector member of the hub of FIG. 6.

[0026] FIG. **9** is a perspective view of an assembly that includes the balloon catheter of FIG. **5A**, first connector member of FIG. **7**, and second connector member of FIG. **8**.

[0027] FIG. **10A** is a perspective view of an exemplary embodiment of a hub housing of the hub of FIG. **6**.

[0028] FIG. **10B** is an end view of the hub housing of FIG. **10A**.

[0029] FIG. **10C** is a perspective view of a first housing portion of the hub housing of FIG. **10A**.

[0030] FIG. **10D** is an elevational view of the first housing portion of FIG. **10C**.

[0031] FIG. **11A** is a perspective view of an exemplary embodiment of a housing ring of the hub of FIG. **6**.

[0032] FIG. **11B** is a cross-sectional view of the housing ring of FIG. **11A** taken along a midline thereof.

[0033] FIG. **12** is a perspective view of an exemplary embodiment of an absorbent member of the nasal catheter assembly of FIG. **2**.

[0034] FIG. **13A** is a perspective view of an exemplary embodiment of a washer of the nasal catheter assembly of FIG. **2**.

[0035] FIG. **13B** is an elevational view of the washer of FIG. **13A**.

[0036] FIG. **14** is a cutaway view of an assembly that includes the absorbent member of FIG. **12** and the washer of FIG. **13A**.

[0037] FIG. **15A** is a perspective view of an exemplary embodiment of a nut of the nasal catheter assembly of FIG. **2**.

[0038] FIG. **15B** is a cross-sectional view of the nut of FIG. **15A** taken along a midline thereof.

[0039] FIGS. **16A-16D** illustrate an exemplary method of using the nasal catheter assembly.

DETAILED DESCRIPTION

[0040] When referring to specific directions in the following discussion of exemplary devices, systems, and assemblies, it should be understood that such directions may be described with regard to the orientation and position of such devices, systems, and assemblies relative to the human body. Thus, the term “anterior” refers to a location closer to the front of the body or the face, and the term “posterior” refers to a location closer to the back of the body. Reference to specific directions may also be made relative to the operator of the exemplary devices, systems, and assemblies disclosed herein. In this regard, the term “proximal” means closer to the operator or in a direction toward the operator, and the term “distal” means more distant from the operator or in a direction away from the operator. Also, it should be understood, as used herein, the term “posterior” is interchangeable with the term “distal,” and the term “anterior” is interchangeable with the term “proximal.”

[0041] FIG. **1** depicts a nasal catheter system **10** according to an embodiment of the present disclosure. System **10** generally includes a fluid delivery device **50** and a nasal catheter assembly **100**. Fluid delivery device **50** may be a syringe with a connector **52** for connecting to nasal catheter assembly **100**. For example, connector **52** of fluid delivery device **50** may be a Luer lock connector which may include a male taper member **54** and a threaded collar **56**. However, other connectors are contemplated, such as a Luer slip connector, snap-fit connectors, barbed connectors, and the like, for example. Fluid delivery device **50** may be configured to transfer fluid to nasal catheter assembly **100** and preferably in metered volumes. Such fluid may include liquids (e.g., saline) and/or gases (e.g., air).

[0042] FIGS. **2-15B** depict nasal catheter assembly **100**. Nasal catheter assembly **100** generally includes a balloon catheter **110**, a hub **102**, an absorbent member **180**, a washer **190**, and a nut **200**.

[0043] FIGS. **3A-5C** depict balloon catheter **110**. Balloon catheter **110** preferably includes two balloons. However, in some embodiments, a single balloon may be provided. In the particular embodiment depicted, balloon catheter **110** includes a first balloon or posterior balloon **112** and a second balloon or anterior balloon **114**. More specifically, balloon catheter **110** may include a stylet **120**, a backbone **130**, and a balloon skin **111**. Posterior balloon **112** may be of a same or different size than anterior balloon **114**. In some examples, posterior balloon **112** may be smaller than

anterior balloon **114**. Posterior balloon **112** may be axially spaced from anterior balloon **114**. [0044] FIGS. **4A-4C** depict stylet **120**. Stylet or first catheter **120** may be an extrusion of rigid polymer material, such as polyetheretherketone (“PEEK”), for example. Stylet **120** may include an outer surface **122** and an inner surface **123** which both may be cylindrical, as shown in FIG. **4C**. Inner surface **123** may define a lumen **121** which may extend along and through the entire length of stylet **120**. Such lumen **121** may also be the lumen for balloon catheter **110** and, as such, may be the main channel delivering fluid to balloons **112** and **114**. Stylet **120** may be divided between a posterior portion and an anterior portion. As shown in FIG. **4A**, the double-headed arrow illustrates the posterior (“P”) and anterior (“A”) directions. Stylet **120** may include a plurality of openings **124** which may extend through a wall of stylet **120** in a direction transverse to a longitudinal axis of stylet **120**. As shown in FIGS. **4A** and **4B**, the anterior portion of stylet **120** may include a first opening **124a** and second opening **124b**, and the posterior portion of stylet **120** may include a third opening **124c** and a fourth opening **124d**. In the embodiment depicted, each of these openings **124a-d** are blind openings in that they extend through the stylet wall and intersect lumen **121**, but do not continue further back through the wall. Additionally, openings **124a-d** are not only offset longitudinally, but may also be offset rotationally. In other words, first opening **124a** may be rotationally offset 90 degrees relative to second opening **124b**, and third opening **124c** may be rotationally offset 90 degrees relative to fourth opening **124d**. Furthermore, first and fourth openings **124a**, **124d** may be offset 180 degrees, and second and third openings **124b**, **124c** may be offset 180 degrees. Thus, none of the openings **124a-d** of stylet **120** may lie on the same side of stylet **120**. Thus, fluid flowing through lumen **121** of stylet **120** not only travels axially through lumen **121** but may also travel transversely in four orthogonal directions through openings **124a-d**. Although four openings **124a-d** are shown, more or less openings are contemplated which may depend on the desired volumetric flow through each portion of stylet **120** and also the length of the stylet **120**.

[0045] FIGS. **5A-5C** depict backbone **130**. Backbone or second catheter **130** may be an extrusion of flexible polymer material, such as silicone, for example. Backbone **130** may include an outer surface **131** and an inner surface **133** which both may be cylindrical, as shown in FIG. **5C**. However, unlike stylet **120**, outer surface **131** and inner surface **133** may each include a plurality of longitudinally extending ribs **135** arrayed about a longitudinal axis of backbone **130**. Such ribs **135** may define longitudinal channels which may direct fluid axially toward opposing ends of backbone **130**. Inner surface **133** may define a lumen **131** which may extend along and through the entire length of backbone **130**. Such lumen **131** may be configured to receive stylet **120**. Backbone **130** may include a plurality of openings **134** which extend through a wall of backbone **130** in a direction transverse to a longitudinal axis of backbone **130**. As shown in FIGS. **5A** and **5B**, the anterior portion of backbone **130** may include a first opening **134a**, a second opening **134b**, and a third opening **134c**, and the posterior portion of backbone **130** may include a fourth opening **134d** and a fifth opening **134e**. In the embodiment depicted, first, second, and third openings **134a-c** may be blind openings and may each lie in a single plane and extend through the same side of backbone **130**. On the other hand, fourth and fifth openings **134d**, **134e** may be through openings in that they may extend through the wall of backbone **130**, intersect lumen **131**, and continue extending back through the wall. Fourth and fifth openings **134d**, **134e** may be rotationally offset by 90 degrees. However, as shown, fifth opening **134e** may be coplanar with openings **134a-c**. Like stylet **120**, more or less openings **134** are contemplated at least depending on the desired volumetric flow and the length of balloon catheter **110**.

[0046] As best shown in FIG. **3B**, a posterior end of stylet **120** extends from a posterior end of backbone **130**, and balloon skin **111** extends over both backbone **130** and stylet **120**. Balloon skin **111** is segmented between first balloon **112** and the second balloon **114**. First balloon **112**, which also defines a tip **113** of balloon catheter **110**, extends over the posterior end of stylet **120** and backbone **130** including third and fourth openings **124c-d** of stylet **120** and fourth and fifth

openings **134d**, **134c** of backbone **130**. As shown, third opening **124c** of stylet **120** is aligned with fifth opening **134e** of backbone **130**. Thus, fluid exiting these openings **124c**, **134e** will tend to expand first balloon **112** eccentrically relative to a longitudinal axis **A1** of balloon catheter **110**. [0047] The second balloon **114** extends over the anterior portions of stylet **120** and backbone **130** including first and second openings **124a-b** of stylet **120** and first, second, and third openings **134a-c** of backbone **130**. However, unlike with first balloon **112**, none of the openings **124a-b**, **134a-c** between stylet **120** and backbone **130** here are aligned. Thus, fluid traveling through first and second openings **124a-b** of stylet **120** meets more resistance than at the posterior portion of balloon catheter **110** and consequently will take longer to exit into second balloon **114** from openings **134a-c** which has a resulting effect of differential inflation of first and second balloons **112**, **114** from a single, main lumen **121**. Such differential inflation may be further facilitated by the posterior end of lumen **121** dispensing directly into first balloon **112**, as best shown in FIG. 3B which, as mentioned above, is in contrast to the indirect route from stylet **120**, along ribs **135**, and through backbone **130** for filling second balloon **114**. Differential thicknesses between balloons **112** and **114** may also be utilized to achieve the desired differential filling. In this regard, the material thickness of first balloon **112** may be thinner than the material thickness of second balloon **114**. Thus, in some embodiments any or all of such features may be utilized in balloon catheter **110** so that the fill rate of first balloon **112** is generally greater than that of second balloon **114** such that first balloon **112** expands faster than second balloon **114** and reaches a fully expanded/filled state prior to second balloon **114**. In such embodiments, second balloon **114** may at some point expand while first balloon **112** is also expanding but at a slower rate. However, in other embodiments, the resistance to filling second balloon **114** may be tailored such that the pressure needed to expand second balloon **114** may only be achieved after first balloon **112** has been expanded to the requisite extent to apply the appropriate pressure to the tissues within the nasal cavity. Thus, in such other embodiments, second balloon **114** may not begin to expand until first balloon **112** is fully expanded at the target location within the nasal cavity. In any event, the differential filling of balloons **112** and **114** results in first balloon **112** being lodged first within the nasal cavity at a target location to treat posterior epistaxis. Moreover, differential filling from a single, main lumen **121** and a single interface **144a** with fluid delivery device **50** simplifies the procedure particularly for medical personnel inexperienced in treating posterior epistaxis.

[0048] Additionally, at least because the openings **134a-c** are each aligned at the same side of backbone **130**, once fluid exits such openings **134a-c**, second balloon **114** will tend to expand eccentrically in the same eccentric direction as first balloon **112**. To further facilitate eccentric expansion, a line of adhesive **115** may be applied between backbone **130** and first and/or second balloons **112**, **114** at a side of catheter **110** opposite the eccentric expansion which has the effect of preventing expansion at the side of adhesive **115**. Although first and second balloons **112**, **114** are described as being inflatable eccentrically, balloon catheter **110** may be configured such that first and second balloons **112**, **114** expand symmetrically and radially outwardly relative to stylet **120** and backbone **130**.

[0049] Also, as shown in FIG. 3C, balloon catheter **110** may be provided with a pre-curve which may be facilitated by the generally rigid stylet **120**. Such pre-curve may define a center of curvature and a radius of curvature **RI** such that balloon catheter **110** has a concave curvature generally matching a convex curvature of a patient's palate anatomy (see FIG. 16A). The rigid structure of balloon catheter **110** coupled with the pre-curve may help insertion of balloon catheter **110** into a patient's nasal cavity. In addition to the pre-curve, the eccentric expansion mentioned above may be configured such that first and second balloons **112**, **114** expand outwardly in a direction away from the center of curvature. This may allow balloon catheter **110** to rest in the desired position on the patient's palate while the balloons **112**, **114** expand outwardly into engagement with the nasal cavity.

[0050] FIGS. 6-11B depict hub **102** of nasal catheter assembly **100**. In the depicted embodiment,

hub **102** generally includes a first connector **140**, a second connector **150**, a hub housing **160**, and a housing ring **170**.

[0051] FIG. 7 depicts first connector **140**. First connector **140** generally includes a connector body **141**, a first connector end **144a**, and a second connector end **144b**. Connector body **141** may be cylindrical and may include a flange **143** extend radially outwardly therefrom at an anterior end thereof. A valve **149** may be disposed within body **141**. Such valve **149** may be a one-way check valve permitting flow from first connector end **144a** to second connector end **144b**, but not vice versa. First connector end **144a** may define an inlet port **147** for fluid introduction into first connector **140** and ultimately into lumen **121** of balloon catheter **110**. First connector end **144a** may be configured to connect to fluid delivery device **50**. In this regard, first connector end **144a** may include a Luer lock connector which may include a female taper opening **147** and a flange **145**. However, other connectors are contemplated so as to correspond with fluid delivery device **50**, such as a Luer taper and the like. Second connector end **144b** may be configured to connect to second connector **150** and, in this regard, may include Luer lock connector which may include a male taper member **146** and threaded collar **148**, as shown. Again, other connectors are contemplated.

[0052] FIG. 8 depicts second connector **150** which may include a body **151**, a first connector end **154a**, and a second connector end **154b**. Second connector **150** may operate as an adapter/coupling between first connector **140** and balloon catheter **110**. In this regard, first connector end **154a** may be configured to connect to second connector end **144b** of first connector **140** and, as such, may include a Luer lock. On the other hand, second connector end **154b** may include a barbed taper **156** for a fluid-tight engagement between second connector **156** end and balloon catheter **110**.

[0053] FIG. 9 depicts an assembly of first and second connectors **140**, **150** and balloon catheter **110**. In this regard, second connector end **144b** of first connector **140** is connected to first connector end **154a** of second connector **150**. Also, second connector end **154b** of second connector **150** is connected to anterior end of balloon catheter **110** such that lumen **121** is in communication with inlet port **147** of first connector **140**.

[0054] FIGS. 10A-10D depict the hub housing **160**. Hub housing **160** may include a first housing portion **161a** and a second housing portion **161b** which may connect together over first and second connectors **140**, **150** to form hub **102**. First housing portion **161a** is shown in FIGS. 10C and 10D and is like second housing portion **161b**. First housing portion **161a** includes a body **162**, a nose cone **163** extending from one end of body **162**, and a neck **167** extending from another end of body **162**. Nose cone **163** may be conically tapered and may include a post **165a** and a post hole **165b** for corresponding engagement with the same features of second housing portion **161b**. Body **162** may be cylindrical and may include a plurality of ridges **166** arrayed along a longitudinal axis thereof and extending transverse relative to the longitudinal axis. Such ridges **166** may be in the form of one or more helical threads. Alternatively, such ridges **166** may be in the form of ratchet teeth. Neck **167** may not include ridges **166** but may include a groove **168**. As mentioned, first housing portion **161a** may be positioned over first and second connectors **140**, **150** and may thereby form a shell or housing thereof. In this regard, first housing portion **161a** may include a first surface **164a** configured to receive body **141** of first connector **140**, a second surface **164b** configured to receive body **151** of second connector **150**, and a third surface **164c** configured to receive an anterior portion of balloon catheter **110**. As such, each of these surfaces **164a-c** may be partially cylindrical and, when first and second housing portions **161a-b** come together, form corresponding cavities or openings for such features. Additionally, when first and second housing portions **161a-b** are connected together, as shown in FIGS. 10A and 10B, threads **166** may be interrupted adjacent to an interface between housing portions **161a-b** and opposing flats **169a-b** may be formed which extend longitudinally along such interface, as best shown in FIGS. 10A and 10B.

[0055] FIGS. 11A and 11B depict the housing ring **170**. Housing ring **170** may be a ring or annulus which defines an opening **172** extending therethrough which may be configured to receive an end

portion of hub **102**, which is formed by the necks **167** of first and second housing portions **161a-b**. Opening **172** may also be configured to receive flange **143** of first connector **140**. Ring **170** may also include a circumferential protrusion **174** extending radially inwardly. Such protrusion **174** may engage groove **168** of hub housing **160** in a snap-fit manner. Such connection between housing ring **170** and hub housing **160** secures first and second housing portions **161a-b** together.

[0056] Although hub **102** has been described herein as including a hub housing **160**, first and second connectors **140**, **150**, and housing ring **170**, it should be understood that in some embodiments, first and second connectors **140**, **150** may be integral such that they form a single connector with the desired connector ends (e.g., ends **144a**, **154b**). In other embodiments, all of the aforementioned components of hub **102** may be integral such that hub **102** forms a single monolithic component that includes connector ends (e.g., ends **144a**, **154b**) and ridges **166** on its exterior.

[0057] FIG. **12** depicts an absorbent member **180** of the nasal catheter assembly **100**. In the embodiment depicted, absorbent member **180** is a conical sponge that has a conically tapered outer surface **181** configured to absorb liquids, such as blood. In some embodiments, outer surface **181** may also include a coating or a covering. For example, outer surface **181** may include a Telfa® lining or covering which resists sticking to blood and nasal mucosa thereby making removal of absorbent member **180** from the patient easier than without the Telfa® lining. In another example, outer surface **181** may include a tranexamic acid (“TXA”) coagulant coating which can help stop bleeding. Absorbent member **180** also includes an inner surface **183** that defines a cavity **182**. Such inner surface **183** may be tapered inwardly from an anterior end to a posterior end thereof, as best shown in FIG. **14**. As such, cavity **182** may be conically tapered. Cavity **182** may extend through the posterior end of absorbent member **180** so as to form an opening **185** configured to slidably receive balloon catheter **110**. Although absorbent member **180** is described as being a sponge, absorbent member **180** can be made from other absorbent material, such as gauze, for example. However, a sponge material is preferable as it highly absorbent, is soft, and generally retains its shape.

[0058] FIGS. **13A** and **13B** depict a washer **190** of the nasal catheter assembly **100**. Washer **190** is generally annular such that it has an opening **191** extending therethrough. In the embodiment depicted, such opening **191** is defined by first and second curved surfaces **198a-b** and first and second flat surfaces **196a-b**, as best shown in FIG. **13B**. First and second curved surfaces **198a-b** are disposed opposite each other and are concavely curved. Such curved surfaces **198a-b** are configured to be positioned over threads **166** of hub **160** and to slide along such threads **166**. In other words, a diameter defined between inner curved surfaces **198a-b** is greater than a major diameter of the threads **166** of hub **160**. Flat surfaces **198a-b** are disposed opposite each other and are configured to be positioned along flats **169a-b** of hub **160**. Thus, when washer **190** is positioned over hub **160**, washer **190** may be axially translatable along hub **160**, but may be prohibited from rotation relative to hub **160** at least due to the arrangement of flats **169a-b**. Washer **190** may also include a tapered outer surface **194** and a curved outer surface **192**. Tapered outer surface **194** may be a conically tapered surface which may have a taper corresponding to the taper of inner surface **183** of absorbent member **180**. In this regard, washer **190** may be positioned within cavity **192** of absorbent member **180** such that tapered outer surface **181** engages tapered inner surface **183** which may have the effect of washer **190** pushing radially outwardly on absorbent member **180**. This may provide support for absorbent member **180**, particularly when driven into a nostril of a patient. Curved outer surface **192** may be convexly curved, may intersect tapered surface **194** and may form at least a portion of an anterior side of washer **190**.

[0059] FIGS. **15A** and **15B** depict a nut **200** of nasal catheter assembly **100**. Nut **200** may be a push nut. In this regard, nut **200** may include an annular body **201** defining an opening **204** extending therethrough. Annular body **201** may include a plurality of fingers or flexible members **202** extending radially inwardly into opening **204**. Such fingers **202** may flex in one direction while

being rigid in another direction. As such, when fingers **202** engage a ridge **166** of hub **160**, pushing nut **200** in a first direction may cause fingers **202** to flex and nut **200** to advance to next ridge **166**. Conversely, pushing nut **200** in a second direction opposite the first direction may be resisted by fingers **202** engaging a ridge **166**. Thus, fingers **202** and ridges **166** may act like a ratchet which can be reversed by rotating nut **200** in a first rotational direction. Also, rather than pushing nut **200** along ridges **202**, nut **200** may be advanced by rotating nut **200** in a second rotational direction. Nut **200** may also include a plurality of tabs or wings **206** which may extend anteriorly from annular body **201** and may facilitate rotation of nut **200**. As shown in FIG. **15B**, nut **200** may further include a concavely curved posterior surface **203**. The concave curvature of surface **203** may correspond to a convex curvature of curved surface **192** of washer **190**. In this regard, convex and concave surfaces **190**, **203** may engage each other to provide uniform axial force distribution during axial translation while facilitating rotation of nut **200** relative to washer **190**.

[0060] Although assembly **100** is depicted and described above as including washer **190**, some embodiments of assembly **100** may not include washer **190**. In this regard, annular body of nut **200** may be positioned at least partially within cavity **182** of absorbent member **180** (see e.g., FIG. **16D**) such that absorbent member **180** is moveable along hub **160** in an anterior-posterior direction as previously described. As such, nut **200** facilitates the translation of absorbent member **180** and its positioning within a nostril of a patient.

[0061] FIGS. **16A-16D** depict a method of using nasal catheter assembly **100**. As illustrated in FIG. **16A**, posterior epistaxis typically originates at a target location **26** within a posterior region of the nasal cavity **20**. Also, as illustrated in FIG. **16A**, the path for reaching the target location **26** is generally curved along the palate **24** (hard and soft) from a designated nostril **22**. The pre-curve of balloon catheter **110** helps conform to this natural curvature such that those with less experience treating posterior epistaxis may find it easier to guide balloon catheter **110** to the target location **26** within the nasal cavity **20** as compared to catheters without a pre-curve. In this regard, balloon catheter **110** may be inserted into a designated nostril **22** with the concave curve of balloon catheter **110** pointing downward or inferiorly so that the concave curve matches the convex curve of the palate **24**, as illustrated in FIGS. **16B** and **16C**.

[0062] Once the posterior end of balloon catheter **110** is properly positioned at the target location **26**, fluid delivery device **50** may be used to deliver fluid through inflation port **147** of hub **102**. Due to the configuration of balloon catheter **110**, as described above, first balloon **112** inflates first thereby providing direct pressure against the target location **26**. First balloon **112** also lodges into the nasopharynx and seals it completely hence stopping any blood from entering the throat. As fluid continues to be delivered to nasal catheter assembly **100** from a single inlet port **147**, the fluid eventually makes its way into second balloon **114** such that it is expanded in the anterior portion of the nasal cavity **20**, as illustrated schematically in FIG. **16C**. The eccentric expansion of first and second balloons **112**, **114** allows catheter **110** to rest on the palate **24** which helps maintain its position relative to the target location **26** as each balloon **112**, **114** is inflated.

[0063] Once the balloons **112**, **114** are sufficiently inflated, absorbent member **160** is driven toward the designated nostril **22**. This may be done by pushing nut **200** axially toward the external nose **28** which slides absorbent portion **180** into the designated nostril **22**. The fingers **202** of nut **200** and ridges **166** of hub **160** prevent the absorbent portion **180** from advancing out of the nostril **22**. Additionally, axial pressure imposed on washer **190** by nut **200** creates corresponding radial and axial pressure from within absorbent member **180** which helps seal the nostril **22** from any leakage and further helps apply pressure from within the nostril on Kiesselbach's plexus, which is the anatomical feature responsible for anterior epistaxis. Thus, assembly **100** can address both posterior and anterior epistaxis, concurrently. When it comes time to remove the assembly **100** from the patient's nasal cavity **20**, nut **200** may be rotated in a first direction which backs up nut **200** allowing washer **190** and absorbent portion to back out of the nostril **22**. Balloon catheter **110** may be cut or otherwise drained to deflate balloons **112**, **114** and remove catheter **110**.

[0064] Although the subject matter disclosed herein has been described with reference to particular embodiments, it is to be understood that these embodiments are merely illustrative of the principles and applications of the present invention. It is therefore to be understood that numerous modifications may be made to the exemplary embodiments and that other arrangements may be devised without departing from the spirit and scope of the present disclosure as defined by the appended claims.

Claims

1. A nasal catheter assembly, comprising: a balloon catheter having a first end, a second end, a lumen extending through the first end toward the second end, and a first balloon in communication with the lumen such that fluid flowing through the lumen fills and expands the first balloon; a hub connected to the first end of the balloon catheter and defining an inlet port in communication with the lumen of the balloon catheter such that fluid introduced through the inlet port is directed into the lumen, the hub having a hub housing defining a plurality of ridges extending outwardly therefrom and transverse to a longitudinal axis of the hub; an absorbent member having an outer surface configured to absorb a liquid and an inner surface defining a cavity; a washer moveably disposed over the hub housing and being disposed within the cavity of the absorbent member such that the washer engages the inner surface of the absorbent member; and a nut movably disposed over the hub housing and having at least one finger sequentially engageable with the plurality of ridges for incrementally moving and securing the washer and absorbent member at any one of a plurality of positions along the longitudinal axis of the housing.
2. The assembly of claim 1, wherein the absorbent member is a conical sponge having an opening extending therethrough and in communication with the cavity, the balloon catheter extending through the opening of the conical sponge.
3. The assembly of claim 1, wherein the inner surface of the absorbent member tapers inwardly from a first end toward a second end thereof, and the washer includes an outer surface that has a taper corresponding to the taper of the inner surface of the absorbent member such that advancing the washer within the cavity of the absorbent member applies an outward radial force on the absorbent member.
4. The assembly of claim 1, wherein the washer includes a convex surface, and the nut includes a concave surface engageable with the convex surface of the washer.
5. The assembly of claim 1, wherein the washer and hub include corresponding flats which permit slidable translation of the washer over the hub and prohibit rotation of the washer relative to the hub.
6. The assembly of claim 1, wherein at least a portion of the nut is disposed within the cavity of the absorbent member and engages the washer.
7. The assembly of claim 1, wherein the at least one finger of the nut is a flexible finger configured to permit axial translation of the nut in a first axial direction and resist axial translation in an opposite second axial direction when engaging any one of the ridges of the plurality of ridges.
8. The assembly of claim 7, wherein the plurality of ridges are defined by at least one thread helically extending about the hub housing, and rotating the nut about the longitudinal axis of the hub housing in a first rotational direction moves the nut in the second axial direction.
9. The assembly of claim 1, wherein the hub further includes a first connector member and a second connector member connected to the first connector member, the first connector member defining the inlet port and including a one-way valve therein, the second connector member defining a connector end connected to the first end of the balloon catheter.
10. The assembly of claim 9, wherein the hub housing includes first and second housing portions disposed over at least a portion of each of the first and second connector members.
11. The assembly of claim 10, wherein the hub further includes a housing ring disposed over at

least a portion of the first and second housing portions and securing the first and second housing portions together.

12. The assembly of claim 1, wherein the balloon catheter further includes a second balloon in communication with the lumen such that fluid flowing through the lumen also fills and expands the second balloon.

13. A nasal catheter assembly, comprising: a balloon catheter having a first end, a second end, a lumen extending through the first end toward the second end, and a first balloon in communication with the lumen such that fluid flowing through the lumen fills and expands the first balloon; a hub connected to the first end of the balloon catheter and defining an inlet port in communication with the lumen of the balloon catheter such that fluid introduced through the inlet port is directed into the lumen, the hub having a hub housing defining a plurality of ridges extending outwardly therefrom and transverse to a longitudinal axis of the hub; an absorbent member having a conically tapered outer surface configured to absorb a liquid and an inner surface defining a cavity; and a nut movably disposed over the hub housing and being at least partially disposed within the cavity of the absorbent member, the nut having at least one finger sequentially engageable with the plurality of ridges for incrementally moving and securing the washer and absorbent member at any one of a plurality of positions along the longitudinal axis of the housing.

14. A nasal catheter assembly, comprising: a balloon catheter having a first catheter defining a lumen, a second catheter disposed over the first catheter, and first and second balloons disposed over the first and second catheters, the first and second catheters each having a plurality of transverse openings extending transverse to respective longitudinal axes thereof and arranged relative to each other such that fluid flowing from the lumen and from the transverse openings fills and expands the first balloon and then the second balloon; a hub connected to the balloon catheter and defining an inlet port in communication with the lumen of the balloon catheter such that fluid introduced through the inlet port is directed into the lumen; and an absorbent member slidably disposed over the hub.

15. The assembly of claim 14, wherein the absorbent member is a conical sponge.

16. The assembly of claim 15, further comprising a washer disposed within the conical sponge, the washer being slidably disposed over the hub and rotationally constrained relative thereto.

17. The assembly of claim 14, wherein the first balloon extends over an end of the first catheter, and the lumen of the first catheter extends through the end.

18. The assembly of claim 14, wherein the second catheter has an outer surface defining a plurality of longitudinally extending ribs, and an inner surface defining a plurality of longitudinally extending ribs, the ribs of the inner surface engaging the first catheter, and the ribs of the outer surface interfacing with the first and second balloons.

19. The assembly of claim 14, wherein at least one pair of transverse openings of the first and second catheters feeding fluid to the first balloon are aligned, and each of the transverse openings of the first and second catheters feeding fluid to the second balloon are non-aligned.

20. The assembly of claim 14, wherein the hub further includes a first connector member and a second connector member connected to the first connector member, the first connector member defining the inlet port and including a one-way valve therein, the second connector member defining a connector end connected to the first end of the balloon catheter.
